

Introduction au labo (Kali + Metasploitable2)

Report

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Module: Computer science and network security

1.introduction :

Objective :

The primary objectives of this practical work were to:

- Configure and verify network connectivity between penetration testing tools and target systems
- Perform network reconnaissance using various discovery techniques
- Conduct port scanning and service enumeration to identify potential attack surfaces
- Interact with vulnerable services in a non-destructive manner
- Capture and analyze network traffic to understand protocol security implications
- Document findings and understand the methodology of ethical hacking

Environment

- Virtualization: VirtualBox / VMware (circle one)
- Kali Linux (attacker platform)
- Metasploitable2 (vulnerable target)
- Network: Host-Only
- Tools used: `ip`, `ifconfig`, `ping`, `netdiscover`, `arp-scan`, `nmap`, `ftp`, `curl`, `tcpdump`, Wireshark

2.Startup & Network Check

Procedure :

1. Start both VMs and ensure network mode is Host-Only.
2. On Kali, run: `ip a \` `uname -a \` `whoami`
3. On Metasploitable2, run: `ifconfig #` or `ip a`

Observations

- Kali IP: 192.168.111.128
- Metasploitable2 IP: 192.168.111.129

Deliverable:

```
root@kali: ~  
File Edit View Search Terminal Help  
root@kali:~# ip q  
Object "q" is unknown, try "ip help".  
root@kali:~# ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000  
    link/ether 00:0c:29:fa:98:82 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.111.128/24 brd 192.168.111.255 scope global dynamic eth0  
        valid_lft 1509sec preferred_lft 1509sec  
    inet6 fe80::20c:29ff:fe9a:9882/64 scope link  
        valid_lft forever preferred_lft forever  
root@kali:~# uname -a  
Linux kali 4.9.0-kali4-amd64 #1 SMP Debian 4.9.30-2kali1 (2017-06-22) x86_64 GNU/Linux  
root@kali:~# whoami  
root  
root@kali:~#
```

```
msfadmin@metasploitable:~$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16436 qdisc noqueue  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000  
    link/ether 00:0c:29:26:b8:9c brd ff:ff:ff:ff:ff:ff  
    inet 192.168.111.129/24 brd 192.168.111.255 scope global eth0  
        valid_lft forever preferred_lft forever  
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000  
    link/ether 00:0c:29:26:b8:a6 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.20.8/24 brd 192.168.20.255 scope global eth1  
        valid_lft forever preferred_lft forever  
msfadmin@metasploitable:~$
```

3.Discovery & Quick Scan :

Procedure

1. Ping the target: `ping -c 4 192.168.111.129`

```
connect: Network is unreachable
root@kali:~#
root@kali:~# ping -c 4 192.168.111.129
PING 192.168.111.129 (192.168.111.129) 56(84) bytes of data.
64 bytes from 192.168.111.129: icmp_seq=1 ttl=64 time=2.30 ms
64 bytes from 192.168.111.129: icmp_seq=2 ttl=64 time=1.12 ms
64 bytes from 192.168.111.129: icmp_seq=3 ttl=64 time=2.13 ms
64 bytes from 192.168.111.129: icmp_seq=4 ttl=64 time=44.5 ms

--- 192.168.111.129 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.122/12.540/44.596/18.513 ms
root@kali:~#
```

2. Perform ARP discovery:

`sudo netdiscover -i <iface> # or sudo arp-scan --localnet`

```
root@kali:~# sudo arp-scan --localnet
Interface: eth0, datalink type: EN10MB (Ethernet)
Starting arp-scan 1.9 with 256 hosts (http://www.nta-monitor.com/tools/arp-scan/)
192.168.111.1 00:50:56:c0:00:01 VMware, Inc.
192.168.111.129 00:0c:29:26:b8:9c VMware, Inc.
192.168.111.129 00:0c:29:26:b8:a6 VMware, Inc. (DUP: 2)
192.168.111.254 00:50:56:f6:92:6a VMware, Inc.

4 packets received by filter, 0 packets dropped by kernel
Ending arp-scan 1.9: 256 hosts scanned in 2.480 seconds (103.23 hosts/sec). 4 re
sponded
```

3. Quick **nmap** scan (first 1000 ports):

`nmap -sC -sV -p 1-1000 192.168.111.129 -oN nmap_quick.txt`

`cptr`

Output summary

- Open ports/services found :
 - ftp 21/tcp — vsftpd
 - ssh 22/tcp — OpenSSH
 - http 80/tcp — Apache httpd
 - smb 445/tcp — Samba

3. Targeted Scanning & Non-destructive Interactions

3.1 Web service enumeration

`nmap -p 80 --script=http-enum 192.168.111.129`

`# or`

`nmap -p 80 -sV --script=http-title 192.168.111.129`

Observation:

```
root@kali:~# nmap -p 80 -sV --script=http-title 192.168.111.129

Starting Nmap 7.50 ( https://nmap.org ) at 2025-10-12 11:13 EDT
Nmap scan report for 192.168.111.129
Host is up (0.0012s latency).

PORT      STATE SERVICE VERSION
80/tcp    open  http    Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_ http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
|_ http-title: Metasploitable2 - Linux
MAC Address: 00:0C:29:26:B8:9C (VMware)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 20.02 seconds
```

3.2 FTP interaction (non-destructive)

1. Connect:

ftp 192.168.111.129

login as: anonymous

run `ls` to list directories

Observation :

```
root@kali:~# ftp 192.168.111.129
Connected to 192.168.111.129.
220 (vsFTPd 2.3.4)
Name (192.168.111.129:root): anonymous
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
200 PORT command successful. Consider using PASV.
150 Here comes the directory listing.
226 Directory send OK.
ftp> █
```

3.3 HTTP interaction (headers and saving homepage)

```
curl -I http://192.168.111.129
```

```
curl http://192.168.111.129/ > index.html
```

Observation :

```
root@kali:~# curl -I http://192.168.111.129
HTTP/1.1 200 OK
Date: Sun, 12 Oct 2025 15:06:26 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Content-Type: text/html
```

4. Quick Capture & Analysis (tcpdump / Wireshark)

Procedure

1. Capture HTTP traffic on Kali:

```
sudo tcpdump -i eth0 port 80 -w tp1_http.pcap
```

Capture for 30–60 seconds while requesting the homepage from the target (via `curl` or browser).

```
root@kali:~# sudo tcpdump -i eth0 port 80 -w tp1_http.pcap
tcpdump: listening on eth0, link-type EN10MB (Ethernet), capture size 262144 bytes
^C36 packets captured
36 packets received by filter
0 packets dropped by kernel
root@kali:~#
```

2. Open `tp1_http.pcap` in Wireshark and apply filter `http`.

tp1_http.pcap						
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help						
http						
No.	Time	Source	Destination	Protocol	Length	Info
4	0.001034	192.168.111.128	192.168.111.129	HTTP	145	GET / HTTP/1.1
8	0.029505	192.168.111.129	192.168.111.128	HTTP	71	HTTP/1.1 200 OK (text/h
16	3.361652	192.168.111.128	192.168.111.129	HTTP	145	GET / HTTP/1.1
20	3.412213	192.168.111.129	192.168.111.128	HTTP	71	HTTP/1.1 200 OK (text/h
28	4.802940	192.168.111.128	192.168.111.129	HTTP	145	GET / HTTP/1.1
32	4.857757	192.168.111.129	192.168.111.128	HTTP	71	HTTP/1.1 200 OK (text/h

> Frame 4: Packet, 145 bytes on wire (1160 bits), 1 > Ethernet II, Src: VMWare_fa:98:82 (00:0c:29:fa:98:82), Dst: 00:0c:29:fa:98:82 > Internet Protocol Version 4, Src: 192.168.111.128, Dst: 192.168.111.129 > Transmission Control Protocol, Src Port: 57424, Dst Port: 80 > Hypertext Transfer Protocol		0000 00 0c 29 26 b8 9c 00 0c 29 fa 98 82 08 00 45 0010 00 83 a7 03 40 00 40 06 33 1f c0 a8 6f 80 c0 0020 6f 81 e0 50 00 50 20 89 20 8e 04 76 0a 78 80 0030 00 e5 60 c8 00 00 01 01 08 0a 00 01 87 8c 00 0040 fa c0 47 45 54 20 2f 20 48 54 54 50 2f 31 2e 0050 0d 0a 48 6f 73 74 3a 20 31 39 32 2e 31 36 38 0060 31 31 31 2e 31 32 39 0d 0a 55 73 65 72 2d 41 0070 65 6e 74 3a 20 63 75 72 6c 2f 37 2e 35 32 2e 0080 0d 0a 41 63 63 65 70 74 3a 20 2a 2f 2a 0d 0a 0090 0a
Hypertext Transfer Protocol: Protocol		Packets: 36 · Displayed: 6 (16.7%) Profile: Default

- Locate a **GET /** request and examine details: headers, cookies, User-Agent, response status, and payload content.

```

✓ GET / HTTP/1.1\r\n
  Request Method: GET
    Request URI: /
    Request Version: HTTP/1.1
  Host: 192.168.111.129\r\n
  User-Agent: curl/7.52.1\r\n
  Accept: */*\r\n

```


5.4. Analysis Questions Answered:

User Agent: curl/8.15.0

Cookies: No cookies present

GET Request: GET / HTTP/1.1 (requesting root page)

6. Conclusion

Summarize the lab results, what you learned, and recommended mitigations for the vulnerabilities observed (disable anonymous FTP, enforce HTTPS, patch services, restrict SMB, use firewalls, etc.).